BOUNDED CONSERVATION WITH PARTIAL IDENTIFICATION ON SYNTHETIC CHAIN NETWORKS: A PREREGISTERED-STYLE HOLDOUT WITH NEGATIVE POINT-ESTIMATE ACCURACY

INDEPENDENT RESEARCHER

Arjun Vijay Prakash
Independent Researcher
arjunv.prakash12345@gmail.com

Manuscript draft packaged 2026-11-11

ABSTRACT

We report a transparent, synthetic benchmark for discrete conservation-based traffic state estimation under missing mainline counts and interval bounds on boundary demand. A deployable estimator fits boundary flows inside a hard conservation polytope with linear inequality observations; partial-identification machinery supplies LP extrema on path-flow aggregates. On an amended final holdout (100 scenarios on 10 seeds $\times$ 10 stress variants; inference frozen at `c15d37e54fe38d63`), the deployable bounded point estimate achieves mean missing-link state RMSE 32.41 vehicles on the 81 scenarios with optimal bounded solutions—worse than masked-mean (6.66) and forward-fill (20.60) imputers on the same subset. Nineteen scenarios yield infeasible polytopes by design. We do not claim algorithmic novelty or state-of-the-art accuracy relative to modern graph neural networks or physics-informed deep learning pipelines, which are not included as baselines here. Post-hoc cluster-bootstrap uncertainty and denominator reconciliation were applied after execution (documented erratum); they were not part of the frozen inference hash.

Keywords traffic state estimation · conservation laws · partial identification · missing data · origin–destination feasibility · reproducibility

1 Introduction

Traffic agencies routinely reconcile loop-detector counts with conservation ideas when lanes fail or detectors drift [1]. Complementary lines estimate origin–destination (OD) tables from partial link volumes via linear programs [6, 5] and related entropy formulations [8], while recent learning methods target missing spatio-temporal fields with graph or physics-informed structures [4, 7]. Set-identification theory clarifies when models identify bounds rather than points [3, 2].

This study is *not* positioned as a new estimator or a conference-level accuracy advance. We document an engineering research stack (`src/research/`) that couples (i) nonnegative LP aggregate bounds on path flows and (ii) a deployable boundary conservation polytope fit with hard observation inequalities. The contribution is methodological transparency: explicit units, stress variants, holdout retirement history, and negative accuracy relative to simple baselines on a single completed final matrix.

2 Problem setup and notation

Time is discretized with fixed step Δt (hours in configuration JSON). For a corridor chain with L links and horizon T steps, let $N \in \mathbb{R}^{L\times T}$ denote link storages (vehicles). Boundary injections are $u \in \mathbb{R}^{2\times T}$ on dedicated inlet and outlet

channels mapped by $R \in \mathbb{R}^{L\times 2}$. Interior path flows z enter through B. The discrete balance enforced in simulation and estimation is

$$N_{:,t+1} = N_{:,t} + \Delta t \left(Bz_{:,t} + Ru_{:,t}\right). \tag{1}$$

Counts on observed mainline cells are represented as intervals $[l, u]$ (vehicles), with a Boolean `count_observed_mask`. Partial identification treats $f \geq 0$ path flows with incidence A such that observed rows satisfy $l \leq Af \leq u$; HiGHS `linprog` returns extrema or explicit infeasibility without midpoint substitution.

3 Deployable estimator versus diagnostics

Deployable default. `baselines.bounded_boundary_estimate` calls `fit_boundary_conservation` with $w_{\text{obs}} = 0$ and hard observation constraints: masked storages enter as linear inequalities in the same polytope used for LP extrema. Boundary channels are pulled toward priors u_{prior} inside the feasible set.

Not claimed. Feasibility of the conservation polytope does not guarantee that the generative truth trajectory lies inside the identification set; LP coverage counts truth witnesses under stated boxes, which differs from net interval coverage over all 100 scenarios. The algebraic conservation residual is a feasibility diagnostic, not evidence that point predictions beat simple imputers.

Oracle and OD NNLS. An oracle boundary channel using generative truth is recorded only for diagnostic RMSE (0 on shared-valid rows by construction). OD NNLS totals in frozen v4 are withdrawn from comparative claims due to an unmasked-incidence implementation bug (erratum); raw JSON values remain audit-only.

4 Synthetic simulator (scope limits)

The generative model is a supply-capped discrete queue on a chain: send/receive caps at each step reject infeasible flux without an external queue. It is not a Cell Transmission Model or LWR field solve; $q = kv$ is not used as independent physics evidence. Synthetic truth is known; measured adapters touch NGSIM I-80 and I-24 samples only as development feasibility (65 s cadence; I-24 units blocked for conservation metrics per project protocol).

5 Evaluation protocol and transparency

Development pilots use seeds 11–55. Holdout generations v1–v3 were retired after exposure or a generative failure; v4 amended reserve seeds are declared in `final_holdout_v4_seed_manifest.json` and executed once (fingerprint `0040a9f70180ada0`). Registry and cooperative freeze receipts deter accidental reuse but are not a cryptographic cross-host security boundary. A post-execution analysis erratum corrected bootstrap multiplicity and paired comparisons without rerunning scenarios (`analysis_script` SHA-256 prefix `8ac2ca5f2b8136d8`). Historical `build_verification.json` still records the pre-correction analysis prefix `e00d4da51c0b965e`; manuscript numerics follow `research_manuscript/packaging_analysis_receipt.json`.

Stress variant `wrong_interval` keeps generative truth inside a stated narrower boundary box (truth-containing interval stress), not an empty misspecified set.

6 Results

Table 1 summarizes matrix-level outcomes. Table 2 reports the declared primary metric `bounded_state_rmse_missing` against baselines. On shared-valid rows ($n = 81$), bounded deployable error exceeds masked mean and forward fill; paired bounded-minus-masked mean difference averages 25.75 vehicles with bootstrap 95% interval [21.17, 31.33] (post-hoc). Table 3 shows that bounded estimates are closer on observed links yet worse when missing links are included—as expected when hard constraints do not align with truth on unobserved cells.

7 Illustrative non-identifiability (not a new theorem)

When only a single aggregate of nonnegative path flows is observed with a wide upper bound, multiple path splits remain feasible; the LP interval for total flow then has positive width even with perfect count sensors on other links. We include this as a textbook-style consequence of partial identification, not as a formal novelty claim.

Table 1: Final holdout v4 matrix outcomes (10 stress variants $\times$ 10 seeds).

Quantity	Count
Scenarios completed	100
Bounded point estimate valid	81
Infeasible conservation polytope	19
Infeasible: truth_outside_interval	10
Infeasible: intentional_boundary_inconsistency	8
Infeasible: gaussian_misspecification	1
Rows with generative boundary truth in stated box	80
Boundary interval covers truth (net vehicles), matrix	80/100
LP data covers truth (bounded-valid rows only)	79/81
LP box covers truth (bounded-valid rows)	79/81

Table 2: State RMSE on missing links (vehicles). Shared-valid denominator n=81 rows with optimal bounded LP point estimate; full-matrix column uses all n=100 scenarios where the metric is finite. Bootstrap: 10 seed clusters, 10,000 replicates (`resampling_seed=424242`); post-hoc, not frozen at preregistration.

Method	Shared valid (n=81)		Full matrix (n=100)	
	Mean	Δ bounded$-$base	Mean	n finite
Bounded deployable LP	32.41	—	32.41	81
Masked mean	6.66	25.75	6.66	100
Forward fill	20.60	11.81	20.67	100
Interval midpoint	66.14	-33.73	71.74	100
Oracle (diagnostic)	0.00	32.41	0.00	100
Bootstrap 95% CI for bounded missing RMSE: [27.85, 37.88]				
Paired diff bounded$-$masked mean: mean 25.72 [21.17, 31.33]				

8 Discussion

Conservation versus accuracy. Enforcing discrete balance and interval observations defines a physically structured feasible set, yet on this synthetic holdout the deployable interior point does not minimize missing-link RMSE relative to masked mean.

Missing strong baselines. Graph Markov networks [4], sparse Bayesian field estimators, and physics-informed learning [7] are absent; any comparison would be speculative.

Protocol amendments. v4 followed a generative repair and failed v3 archive; it is not a blind repeat of the original v3 preregistration.

9 Limitations and future work

Small chain geometry, a single completed final matrix, seed-cluster bootstrap with only ten clusters, and no independent external OD truth limit external validity. Fixing OD NNLS masking is documented but not applied to frozen v4. Embargoed real-network adapters remain future work.

Declarations

Author identity and contact appear on the title page. See companion Markdown `AUTHOR_VERIFICATION_REQUIRED.md` for funding, ethics, and CRediT attestations still requiring human confirmation. AI assistance was used for manuscript packaging and literature verification; human authors remain responsible for contributions and submission ethics.

Table 3: RMSE slices on shared-valid rows (n=81): missing, observed, and all links.

Method	Missing	Observed	All
Bounded deployable	32.41	10.46	17.75
Masked mean	6.66	7.45	7.41
Forward fill	20.60	7.45	11.93

Table 4: Per-variant deployable primary metric `bounded_state_rmse_missing` (conditional mean, valid bounded rows only).

Variant	n valid	n infeasible	Mean RMSE	Std
baseline	10	0	27.29	9.17
contiguous_dropout	10	0	35.55	3.15
gaussian_misspecification	9	1	42.29	14.56
inconsistent_od_counts	10	0	27.29	9.17
intentional_boundary_inconsistency	2	8	46.33	1.83
od_prior_shift	10	0	29.93	10.68
random_missing	10	0	38.19	12.80
shifted_demand	10	0	29.14	8.17
truth_outside_interval	0	10	—	—
wrong_interval	10	0	27.82	8.85

Data and code availability

Synthetic artifacts: `results/research_final_v4/`. Table and figure regeneration: `research_manuscript/scripts/generate_manuscript_tables.py`, `make_manuscript_figures.py`. Immutable input verification: `research_manuscript/scripts/verify_immutable_inputs.py`. Layout: `arxiv.sty` from George Kour's MIT-licensed arxiv-style (see `TEMPLATE_PROVENANCE.md`).

References

[1] Pablo Cáceres, William J. Wright, Roberto Horowitz, and Pravin Varaiya. Freeway loop detector data reconciliation based on vehicle conservation. *Procedia Computer Science*, 151:702–709, 2019.

[2] Alfred Galichon and Marc Henry. Set identification in models with multiple equilibria. *The Review of Economic Studies*, 78(4):1192–1216, 2011.

[3] Francesca Molinari. Microeconometrics with partial identification. *Annual Review of Economics*, 12:355–377, 2020.

[4] Ziyuan Pu, Han Shen, Ruilin Liu, Xiaoliang Wu, Sami Asmar, and Roberto Horowitz. Graph markov network for traffic forecasting with missing data. *Transportation Research Part C: Emerging Technologies*, 117:102671, 2020.

[5] Hanif D. Sherali, Arvind Narayanan, and R. Sivanandan. Estimation of origin–destination trip-tables based on a partial set of traffic link volumes. *Transportation Research Part B: Methodological*, 37(9):815–836, 2003.

[6] Hanif D. Sherali, R. Sivanandan, and Antoine G. Hobeika. A linear programming approach for synthesizing origin-destination trip tables from link traffic volumes. *Transportation Research Part B: Methodological*, 28(2):93–118, 1994.

[7] Rongye Shi, Zhaobin Mo, Kuang Huang, Xuan Di, and Qiang Du. A physics-informed deep learning paradigm for traffic state and fundamental diagram estimation. *IEEE Transactions on Intelligent Transportation Systems*, 23(7):8574–8586, 2022.

[8] Chi Xie, Kara Maria Kockelman, and Steven Travis Waller. A maximum entropy-least squares estimator for elastic origin–destination trip matrix estimation. *Transportation Research Part B: Methodological*, 45(9):1465–1482, 2011.

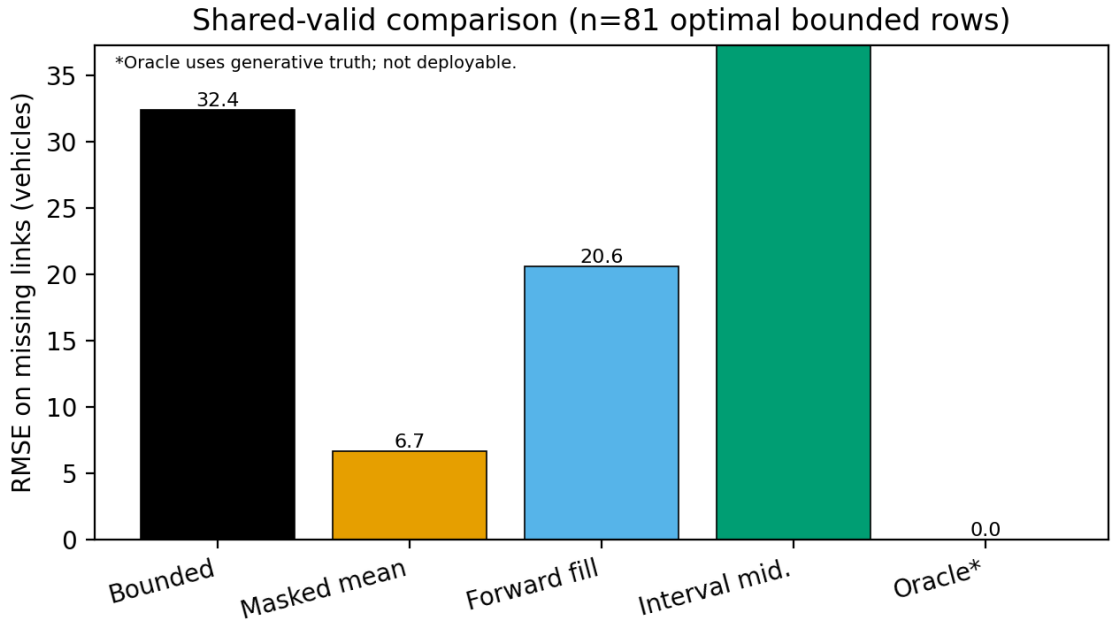


Figure 1: Shared-valid missing-link RMSE ($n = 81$). Oracle bar uses generative truth and is not deployable.

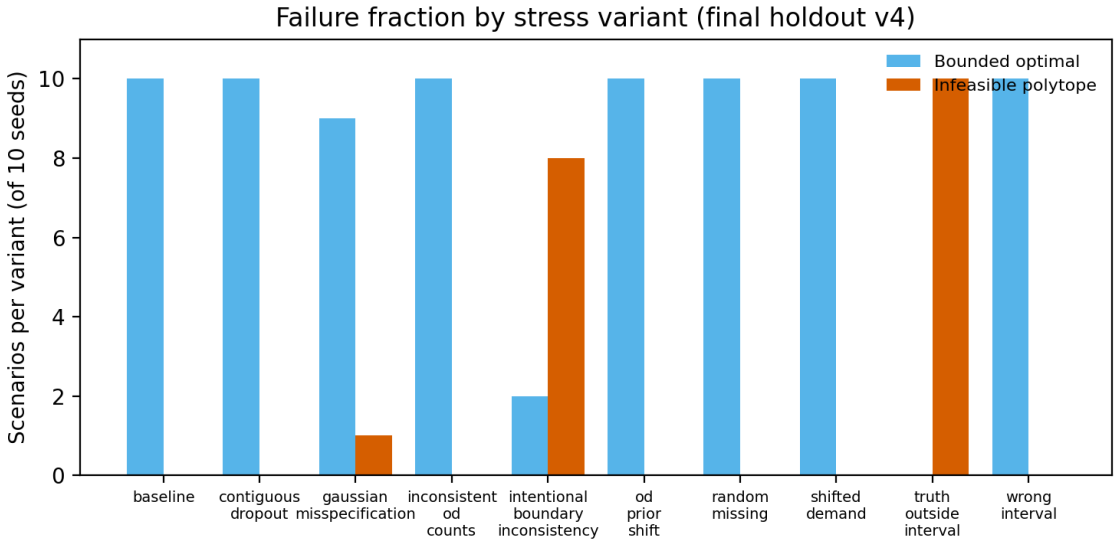


Figure 2: Counts of optimal bounded solutions versus infeasible polytopes per stress variant (10 seeds each).